INCH-POUND
MIL-R-24049B
14 June 1991
SUPERSEDING
MIL-R-24049A
28 March 1966
(See 6.10)

MILITARY SPECIFICATION

ROPE, POLYPROPYLENE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of polypropylene rope.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-C-3131 - Cordage; Packaging of.
MIL-L-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4020
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.
FED-STD-595 - Colors Used in Government Procurement.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 885 - Standard Method of Testing Tire Cords, Tire Cord Fabrics, and Industrial Filament Yarns Made from Man-Made Organic-Base Fibers.
D 1141 - Standard Specification for Substitute Ocean Water. (DoD adopted)
D 1525 - Standard Test Method for Vicat Softening Temperature of Plastics. (DoD adopted)
D 1577 - Standard Test Methods for Linear Density of Textile Fibers.
D 2257 - Standard Test Method for Extractable Matter in Textiles. (DoD adopted)
D 2258 - Standard Practice for Sampling Yarns for Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Material. The material used in the rope shall be virgin, continuous monofilament polypropylene fiber of at least 6.0 grams per denier tenacity and shall range in size from 100 to 600 denier per filament (see 6.3). The material shall have a softening point of 300 degrees Fahrenheit (°F) and a specific gravity not greater than 0.91. All fibers shall contain sufficient heat stabilizers to meet the softening point requirements when tested as specified in 4.6.12.

Source: https://assist.dla.mil - Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

3.2.1 Ultraviolet resistance. The polypropylene fibers shall contain sufficient ultraviolet stabilizers to meet the requirements when tested as specified in 4.6.11. The specimens removed from the rope shall have a breaking strength loss not greater than 50 percent when compared with the original specimens.

3.3 Construction. The ropes shall be of three strands. Each strand shall be made of one size yarn and shall have a sufficient number of yarns to produce a rope conforming to the requirements of this specification (see 6.3). The lay of the rope shall be right-hand or "Z" lay.

3.3.1 Physical properties. The physical properties of the finished rope shall be as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

TABLE I. Physical properties.

Circumference at load P (inches)	Tolerance plus or minus (inch)	Approximate diameter (inches)	Load P 200 x D^2 (pounds)	Hardness min. (pounds)	Linear density +5 percent -0 percent pounds/100 ft.	Breaking strength (min) (pounds)
5/8	1/16	3/16	7	5	.84	720
3/4	1/16	1/4	12	5	1.19	1,130
1	1/16	5/16	20	5	2.12	1,710
1-1/8	1/16	3/8	28	5	2.77	2,440
1-1/4	1/16	7/16	38	5	3.33	3,160
1-1/2	1/16	1/2	50	5	4.76	3,780
1-3/4	1/16	9/16	65	5	6.32	4,600
2	1/8	5/8	80	5	8.33	5,600
2-1/4	1/8	3/4	110	5	10.98	7,650
2-1/2	1/8	13/16	130	5	13.15	8,900
2-3/4	1/8	7/8	153	5	15.74	10,400
3	3/16	1	200	20	19.23	12,600
3-1/2	3/16	1-1/8	250	20	26.31	16,500
3-3/4	3/16	1-1/4	310	20	30.30	18,900
4	3/16	1-5/16	345	20	34.24	21,200
4-1/2	1/4	1-1/2	450	20	43.85	26,800
5	1/4	1-5/8	530	20	52.63	32,400
5-1/2	1/4	1-3/4	610	20	63.29	38,800
6	5/16	2	800	20	79.36	46,800
6-1/2	5/16	2-1/8	900	20	90.90	55,000
7	3/8	2-1/4	1,000	20	111.11	62,000
8	3/8	2-5/8	1,400	20	142.85	81,000
9	7/16	3	1,800	20	183.48	103,000
10	7/16	3-1/4	2,100	20	232.55	123,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

3.3.1.1 Elongation. The elongation at the breaking strength shall not be greater than 45 percent. The load elongation curve drawn autographically shall not exhibit evidence of sudden changes in load applications greater than 5 percent of the load weighed at the instant of change. Changes due to splice slippage shall not be considered in this determination.

3.3.1.2 Hardness. The furnished rope shall meet the minimum hardness specified in table I (see 4.6.13).

3.3.2 Identification ticket. Each reel or spool of rope shall have a ticket, identification tag, or label attached to it. The ticket shall be in accordance with type B, class 1, size 4 or 5, 15 CSU grade designator of UU-T-81. The ticket shall be legibly printed, stamped, or typed with water insoluble ink. The ticket shall contain the following information:

Stock number
Nomenclature
Specification
Length
Contract number and date
Year of manufacture
Contractor's name

3.3.3 Identification marker. The manufacturer shall identify his product by inserting a kraft paper or water repellent cotton marker within one strand and completely enveloped by the cover yarns of that strand in all ropes larger than 1-1/4 inches circumference. Unless otherwise specified (see 6.2), the manufacturer's name, year of manufacture, and type of fiber (Polypropylene) shall be printed on the marker in bold, easily-read type. Italic or script type shall not be used. The printing shall not be affected upon exposure to water or mineral oil when tested.

3.3.4 Extraneous material. No extraneous material shall be added for the purpose of weighing the rope, except coloring pigments which are permissible. The extractable matter shall not be greater than 3.0 percent (see 4.6.6).

3.3.5 Moisture content. The moisture content of the rope shall not be greater than 1.0 (see 4.6.9).

3.3.6 Heat aging. The finishing materials shall be resistant to aging and shall cause rope strength loss not greater than 10 percent when tested (see 4.6.10).

3.3.7 Spliceability. The finished ropes shall be spliceable and shall not develop yarn displacement or strand cockles (distortion formed by a back twist in the strand) in the splicing test (see 4.6.7).

3.3.8 Color. Unless otherwise specified, the color of the finished rope shall be black. When colored rope other than black is specified (see 6.2). Pigments shall be added to the polymer at the extruder to give the suitable color in accordance with FED-STD-595.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

3.3.9 Matching. The shade of the finished rope shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500 degrees Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.10 Colorfastness. The finished rope shall show fastness to weathering equal to or better than the standard sample when tested as specified in 4.6. When no standard sample has been established, the colored rope shall show good fastness to weathering when tested in accordance with FED-STD-191.

3.3.11 Put-up. Unless otherwise specified, the rope shall be furnished on non returnable reels not larger than 6 feet in diameter in the lengths specified in table II. Broken lengths will be permitted, but no piece shall be less than 600 feet when measured in the relaxed condition (see 6.2). The ends of all rope shall be cut off squarely and be securely whipped or taped to prevent untwisting or fraying. The reels shall be wound so that each turn and layer is free from entanglement. A plus tolerance of 10.0 percent shall be allowed on the length specified per reel or spool.

TABLE II. Put up.

Circumference (inches)	Minimum length (feet)	Weight per reel (pounds)
5/8	2,250	20
3/4	2,250	28
1	2,250	50
1-1/8	1,620	47
1-1/4	1,200	42
1-1/2	1,200	60
1-3/4	1,200	80
2	1,200	105
2-1/4	1,200	138
2-1/2	1,200	165
2-3/4	1,200	198
3	1,200	242
3-1/2	1,200	332
3-3/4	600	191
4	600	216
4-1/2	600	276
5	600	332
5-1/2	600	389
6	600	500
6-1/2	600	563
7	600	700
8	600	900
9	600	1,156
10	600	1,465

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations specified in 4.5.1 and the tests specified in 4.6.

4.3.1 First article sample. The first article sample for ropes shall be a continuous finished length of 70 feet.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations specified in 4.5 and the tests specified in 4.4.1 and 4.6 (see 6.3).

4.4.1 Material and component inspection. Determination shall be made for all characteristics specified in table III, except material. The linear density per filament and the tenacity of the fiber shall be determined in accordance with ASTM D 885, D 1577 and D 2258. Results shall be either pass or fail.

4.4.2 Lot. For purpose of inspection sampling a lot shall consist of all reels of rope produced in one facility, using the same materials and production processes, and being offered for delivery at one time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

TABLE III. Component testing.

Characteristics	Requirement
Material	3.2
Denier per filament	3.2
Tenacity	3.2
Direction of twist (strand)	3.3

4.5 Examinations.

4.5.1 Examination of the end item for visual defects. A lot size for examination for visual defects specified in table IV shall be expressed in units of reels, each. Ten percent of the gross length contained on each sample unit, but not less than 100 feet shall be subjected to the visual examination. The inspection level shall be as specified in table VI. Any sample unit having one or more defects specified in table IV shall be rejected. If any defects are noted in the original sample units, additional units shall be randomly selected and if any defects specified in table IV are noted the entire lot shall be rejected.

TABLE IV. Visual examination defects.

Examine	Defect
Appearance and workmanship	Cut, chafed or damaged, affecting serviceability
	Kinks, darting yarns 1/ broken or loose ends, bulged strands, strand knots
	Other than 3 strands
	Ends not cut off squarely
	Ends not securely whipped or taped
	Projecting ends
Identification marker 2/	Omitted incorrect illegible
	Italic or script type used

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

TABLE IV. Visual examination defects - Continued.

Examine	Defect
	Not completely covered by cover yarns
	Not as specified
Identification ticket label	Omitted, incorrect, illegible, insecurely attached
	Not as specified
	Handwritten entries
Color	Not as specified
	Specific shade, not within established tolerances
Cleanness	Spot, or stain, clearly noticeable 3/
	Objectionable odor

1/ Darting yarns are internal yarns which project through the cover yarns of the strand at intervals along the rope.
2/ Identification marker for larger than 1-1/4 inch circumference.
3/ At normal inspection distance (approximately 3 feet).

4.5.2 Examination for length and winding. The sample unit for this examination shall be one reel. The inspection level shall be as specified in table V, and the presence of any defect shall be cause for rejection of the entire lot. The lot size shall be the number of units in the inspection lot. Defects shall be as listed in 4.5.2.1 and 4.5.2.2.

TABLE V. Examination for length and winding.

Lot size	Sample size
1-10	All
11-500	10
501-3200	13
3201-10000	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

4.5.2.1 Defects in length. Defects with regard to length shall be considered to exist if any of the following are determined during inspection.

(a) Length of unit less than length specified.
(b) Length of unit less than marked on ticket.
(c) Rope not in continuous length when specified.
(d) Any piece on reels or spools less than 600 feet in length.

4.5.2.2 Defects in winding. Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

(a) Improperly or not firmly wound resulting in kinks, knots, entangling or slippage during unwinding or otherwise affecting free unhampered unwinding of rope.
(b) Knot or splice or otherwise joining of ends to make a continuous length.

4.6 Testing of the end item. Testing of the end item shall be as specified in table VII and 4.6.2 through 4.6.13. The physical and chemical values specified herein apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The quality conformance sample size shall be as specified in table VI and the presence of any defect shall be cause for rejection of the entire lot. The lot size shall be expressed in units of reels, spools, or coils and the sample unit for testing shall be 70 feet of rope.

TABLE VI. Sampling for quality conformance tests.

Number of reels in lot	Number of samples
15 and under	2
16 to 25	3
26 to 90	5
91 to 150	8
151 to 280	13
281 to 500	20
501 to 1200	32
1201 to 3200	50
3201 to 10000	80

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

TABLE VII. Test methods. 1/

Characteristic	Specification reference		No. of determinations per individual sample unit	Results determined as
	Requirement	Test		
Material	3.2		1	Pass or fail
Strand	3.3	Visual	1	Pass or fail
Yarn	3.3	Visual	1	Pass or fail
Lay of rope	3.3	Visual	1	Pass or fail
Circumference	3.3.1	4.6.2	3	Avg. of 3 determinations to nearest 1/16 inch
Linear density	3.3.1	4.6.3	1	See 4.6.3.1
Breaking strength	3.3.1	4.6.4	2	Avg. of 3 determinations to an accuracy of 1 percent
Elongation	3.3.1.1	4.6.5	3	Avg. of 3 determinations to nearest 0.1 percent
Hardness	3.3.1.2	4.6.13	3	Avg. of 3 determinations to nearest 1 pound
Extractable matter	3.3.4	4.6.6	2	Avg. of 2 determinations to nearest 0.1 percent
Moisture content	3.3.5	4.6.9	2	Avg. of 2 determinations to nearest 0.1 percent
Spliceability	3.3.7	4.6.7	1	Pass or fail
Color matching	3.3.9	FED-STD-595	1	Pass or fail
Colorfastness to weathering	3.3.10	Method 5671 of FED-STD-191	1	Pass or fail

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

TABLE VII. Test methods 1/ - Continued.

Characteristic	Specification reference		No. of determinations per individual sample unit	Results determined as
	Requirement	Test		
Identification marker	3.3.3	4.6.8	3	Pass or fail
Softening point	3.2	4.6.12	1	Pass or fail
Heat aging	3.3.6	4.6.10	3	Pass or fail
Ultraviolet resistance	3.2.1	4.6.11	3	Pass or fail

1/ Tests to determine compliance with specification requirements including quantity of delivery may be made under prevailing atmospheric conditions except in settlement of dispute in which case the tests shall be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.

4.6.1 Specimen preparation. The designated length and the number of test specimens shall be removed from the selected test reels, in accordance with the specified test method.

4.6.2 Circumference. The circumference shall be measured in accordance with Method 6003 of FED-STD-191. The results shall be the average of three determinations to the nearest 1/16 inch.

4.6.3 Linear density. The linear density shall be determined in accordance with Method 6004 of FED-STD-191. Tests to determine compliance with the requirements (including quantity for delivery) specified herein, may be conducted under prevailing atmospheric conditions. In case of dispute, the tests shall be conducted on material which has reached equilibrium under standard conditions in accordance with Method 6004 of FED-STD-191.

4.6.3.1 Accuracy. Linear density results shall be determined as follows:

Circumference (inches)	Degree of accuracy (pounds per 100 feet)
5/8 - 1-3/4	Nearest 0.01
2 - 6	Nearest 0.10
6 1/2 - 10	Nearest 1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

4.6.4 Breaking strength. Breaking strength determinations shall be conducted in accordance with Method 6015 of FED-STD-191. Results shall be the average to an accuracy of 1 percent. The load applied need not cause a failure provided all individual test values are greater than the specified minimum breaking strength.

4.6.5 Elongation. The elongation at the minimum breaking strength shall be determined in accordance with the initial single loading procedure of Method 6015 of FED-STD-191. The results shall be the average of the determinations to the nearest 0.1 percent, but no single breaking strength test below the minimum values listed in table I shall be considered acceptable.

4.6.6 Extractable matter. The extractable content shall be in accordance with ASTM D 2257, using a soxhlet apparatus in the procedure. Results shall be the average of two determinations to the nearest 0.1 percent.

4.6.7 Spliceability. A breaking strength specimen shall be prepared and spliced at each end with four full tucks. The specimen shall then be allowed to rest for 24 hours. Three of the tucks shall then be backed out of the rope. That portion of the rope from which the tucks are removed shall be examined for yarn displacement and strand cockles.

4.6.8 Identification marker. Three lengths of the identification marker shall be used to determine the fastness of the printed matter on the identification marker to saltwater and mineral oil. Each length shall be approximately 1-1/2 feet. One length shall be retained as a control, length shall be immersed for 2 hours in synthetic seawater conforming to ASTM D 1141, stock solution no. 1, 2 or 3, and one length shall be soaked for 2 hours in mineral oil conforming to symbol 2190-TEP of MIL-L-17331. Following removal from the respective environments, the two exposed specimens shall be visually compared with the control specimen. The fastness of the printed matter shall be considered satisfactory when no perceptible change in color or legibility is observed.

4.6.9 Moisture content. The moisture content shall be determined in accordance with method 2600 of FED-STD-191. Results shall be the average of two determinations to the nearest 0.1 percent (see 6.5 adjustement for high moisture content).

4.6.10 Heat aging test. Three specimens shall be heated for 5 days in a convection air oven at 175 ± 2°F. The specimens shall then be removed from the oven and conditioned for 24 hours in a standard atmosphere for textiles before being tested for strength. The percent change shall be calculated based on the average strength found for 3 unheated specimens in accordance with method 4100 of FED-STD-191.

4.6.11 Ultraviolet resistance test. Three specimens shall be exposed in a weathering device in accordance with method 5804.1 of FED-STD-191 for 500 hours. The percent change in strength shall be calculated on the average strength found for three unexposed specimens in accordance with method 4100 of FED-STD-191.

4.6.12 Determination of softening point. The softening point of individual fibers, withdrawn from the strands of the rope shall be determined in accordance with ASTM D 1525.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

4.6.13 Hardness. The length of rope previously used in the weight determination shall be taped off at each end. The tip of the 14-inch marlinspike, in accordance with method 6020 of FED-STD-191, shall be inserted between the strands until visible on the opposite side. With the spike inserted, the rope shall be placed in a compression type testing machine in such a manner that the force necessary to push the spike through the rope will be measured with the rope in a relaxed state without tension and completely free to absorb the force of the penetrating spike. The rate of loading shall be 6 ± 1 inch per minute. Care shall be taken to ensure that the spike shall be inserted not less than 5 feet from the end and not less than 4 feet from an area which has been subjected to a previous hardness test. The load necessary to force (to the nearest 1 pound) the spike to the 1/2-inch diameter mark shall be measured on ropes up to and including 2-3/4 inches circumference and to the 1-inch diameter mark for larger size ropes. The marks shall be considered to be reached when the respective mark disappears between the strands.

4.7 Inspection of packaging. Sample packages and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

(a) Treated lumber and plywood. When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container, reel, and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated materials in accordance with MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of interior (unit and intermediate) and exterior fiberboard boxes incuding interior packaging forms shall conform to the class-domestic/fire retardant or class weather resistant/fire retardant materials requirements as specified (see 6.2), of PPP-F-320 and amendments thereto.

5.2 Packaging requirements. The packaging (preservation, packing and marking) requirements shall be in accordance with MIL-C-3131 for the level (A, C, or commercial) of preservation level (A, B, C, or commercial) of packing, and marking including packaging acquisitioning options as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rope covered by this specification is intended for use where light weight and floatability are required. Polypropylene rope made to this specification should not be used with weight handling equipment (overhead lifting) or when it will be exposed to sunlight for extended periods.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) When first article inspection is required (see 3.1).
(d) Circumference size required (see 3.3.1).
(e) When information on identification marker is other than specified (see 3.3.3).
(f) Specific color (shade) required if other than specified (see 3.3.8).
(g) Put-up and length when other than specified (see 3.3.11).
(h) When multiple lengths per unit are not allowed (see 3.3.11).
(i) When fire retardant treated lumber and plywood is required (see 5.1.1 (a)).
(j) Level of preservation, level of packing and other acquisitioning options required (see 5.2).
(k) That the purchaser will accept at original weight any unit which has been shortened or cut for test specimens, if in complete compliance with this specification. Reels should be marked as to actual length.
(l) That polypropylene rope will be purchased on a price-per-pound basis-net weight.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2 and 3.3	DI-MISC-80678	Certification/ data report	10.3.2
4.4	DI-T-2072	Reports, test	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Adjustment for high moisture content. Material furnished containing an excess of moisture will be accepted by an adjustment in weight to the 2 percent moisture basis.

6.6 Certification/data report. When a certification/data report is prepared, the following information should be included:

(a) Conformance of material characteristics to the requirements specified herein (see 3.2).
(b) Conformance to the requirements for finished rope (see 3.3).

6.7 Government or supplier purchases. The requirements specified in 3.3.2 and 3.3.3 apply only to polypropylene rope purchased directly by the Government. All other requirements apply to polypropylene rope purchased as a component for an end item by a supplier and to polypropylene rope purchased directly by the Government.

6.8 Subject term (key word) listing.

Coils
Splice
Spool
Strand cockles
Yarns

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - SH
Air Force - 69

Review activities:
Air Force - 99

User activities:
Army - GL, MO

Preparing activity:
Navy - SH
(Project 4020-0334)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-24049B	14 JUNE 1991

3. DOCUMENT TITLE
ROPE, POLYPROPYLENE

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. Jack Hall (SEA 56W23) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON TPOC: 703-602-1844 332-1844
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:52Z
Check the source to verify that this is the current version before use.